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| **Fiscal Year:** | FY 2013 | **Task Last Updated:** | FY 08/02/2012 |
| **PI Name:** | Lorenzi, Hernan Ph.D. | | |
| **Project Title:** | Study of the Impact of Long-term Space Travel on the Astronaut's Microbiome | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | HUMAN RESEARCH | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **HHC**:Human Health Countermeasures | | |
| **Human Research Program Risks:** | (1) **Medical Conditions**:Risk of Adverse Health Outcomes and Decrements in Performance Due to Medical Conditions that occur in Mission, as well as Long Term Health Outcomes Due to Mission Exposures<br>(2) **Microhost**:Risk of Adverse Health Effects Due to Host-Microorganism Interactions | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | hernan.lorenzi@nih.gov | **Fax:** | FY |
| **PI Organization Type:** | NON-PROFIT | **Phone:** | 301-480-0648 |
| **Organization Name:** | National Institute of Health | | |
| **PI Address 1:** | Laboratory of Biochemistry and Genetics, Cell Cycle Regulation and Nuclear Structure Section | | |
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| **City:** | Bethesda | **State:** | MD |
| **Zip Code:** | 20892 | **Congressional District:** | 8 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 2010 Crew Health NNJ10ZSA003N |
| **Start Date:** | 10/01/2011 | **End Date:** | 09/30/2014 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA ARC |
| **Contact Monitor:** | Griko, Yuri | **Contact Phone:** | 650-604-0519 |
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| **Flight Program:** | ISS | | |
| **Flight Assignment:** | ISS | | |
| **Key Personnel Changes/Previous PI:** | August 2012: Scott Peterson (former co-PI of the project) and Shannon Williamson (key personnel) are not participating in this project any more. Drs. Mark Ott and Duane Pierson are collaborators on this project. | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | NNX12AB02G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | Our goal is to determine how the composition of the human microbiome changes during long-term space exploration and to evaluate its potential impact on astronauts' health. Some microbial species from the human microbiome have a beneficial or protective effect on health; the loss of these species can lead to an altered metabolic function and, in conjunction with reduced immune response, may increase the chance of infection by opportunistic pathogens. In our proposal we will elaborate the notion of the microbiome as harbingers or sentinels to monitor a variety of aspects of the human host, including associations with health status, environmental stress, and exposure to space conditions. By sampling the microbiome of astronauts on earth while in peak physical health and during subsequent times of stress, including long-term exposure to microgravity, g-forces, radiation and changes in health status, we will be able to define signatures of human response to a variety of relevant aspects of space travel. We propose to characterize the bacterial and viral microbiome from various body sites of up to nine astronauts who travel to space at several time points before, during, and after a space mission. Also we will assess the astronauts' immune function before, during, and after the mission by analyzing their collected saliva samples for reactivated latent viruses and cortisol levels, two indicators commonly evaluated during spaceflight immune and stress studies and cytokines from blood samples. Finally, we will correlate the collected microbiome and immune function data with other measured metadata including astronaut health and hygiene as well as environmental factors such as temperature, humidity and environmental microbial samples that will be collected, depending upon availability, from various surfaces on the International Space Station. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The results of this study will provide insights into how the microbial population of the environment affects the composition and dynamics of the human microbiome. This is relevant to studies of respiratory diseases such as asthma and allergies.<br>Investigating the impact of stress and status of the immune system on the human microbiome, and potentially on human health, during a space mission is also applicable to equivalent stressful situations on Earth. Some of the conclusions of this project will also be useful in situations where a group of individuals are confined in a relatively small and closed space for a long period of time, such as a submarine crew. |
| **Task Progress:** | During the first year of this project we have been working on the implementation of different sampling methodologies in space. We have also prepared all the documentation required for starting the recruitment of volunteer astronauts for this study. |
| **Bibliography Type:** | Description: (Last Updated: 01/12/2026) |
| **Abstracts for Journals and Proceedings** | Peterson SN, Torralba M, Ott CM, Pierson DL, Lorenzi HA. "Study of the impact of long-term space travel on the astronauts' microbiome." 2012 NASA Human Research Program Investigators' Workshop, Houston, TX, February 14-16, 2012.<br>2012 NASA Human Research Program Investigators' Workshop, Houston, TX, February 14-16, 2012. , Feb-2012 |